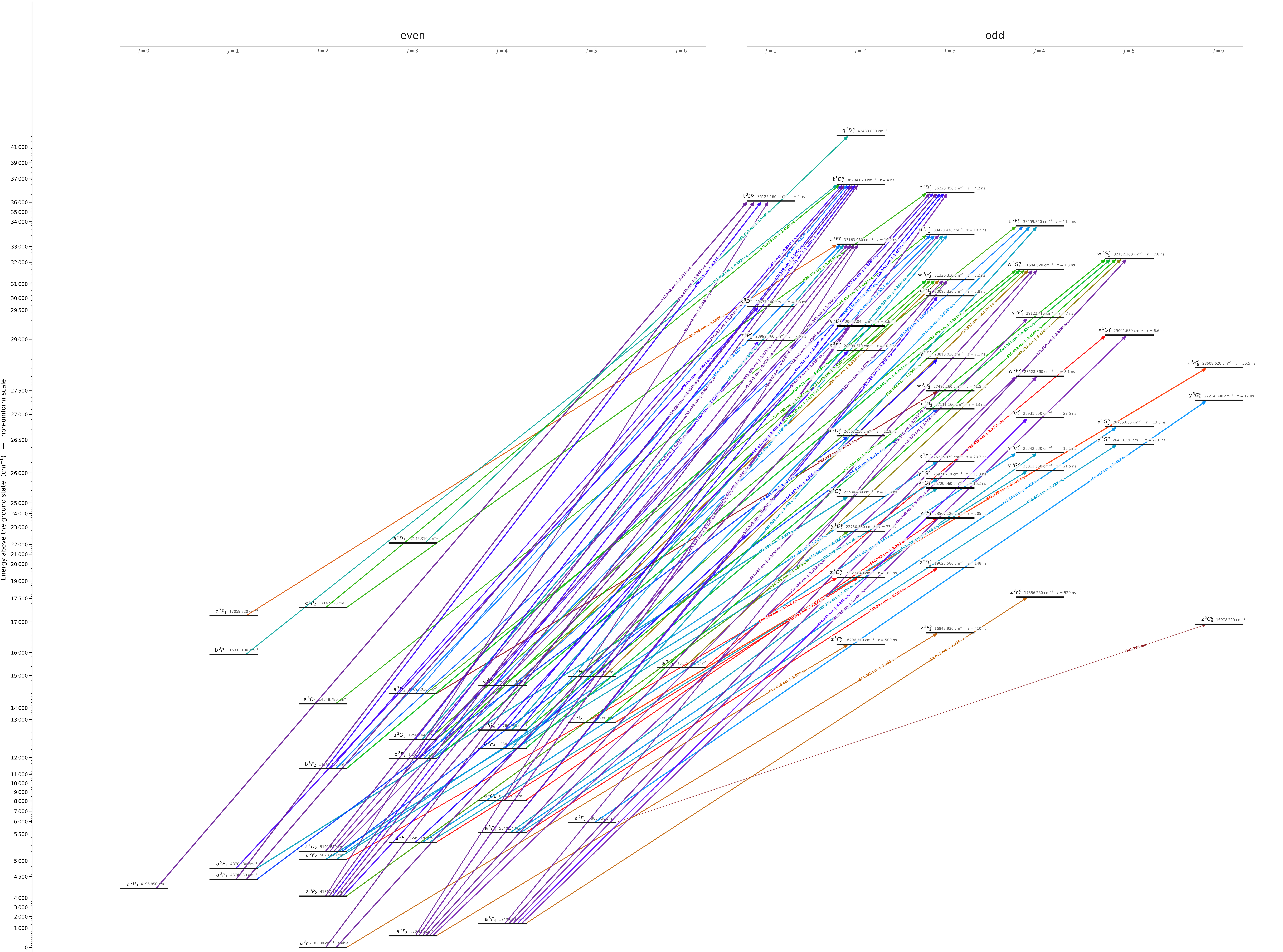
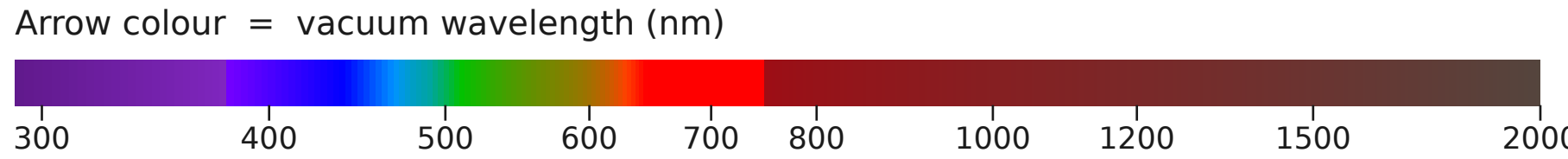


Zirconium

Zr (Zr I). Strongest classified transitions below 2 μm (NIST): 69 levels, 90 lines.

Zr⁺ ionisation limit 53507.83 cm⁻¹ = 6.63413 eV (186.889 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |(U|er|U')| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	(U er U') (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ³ F ₅ – y ² G ₆ ^o	468.9121	468.7809	639.3362	7.423	8.330e+07	13.26	100	proposed laser-cooling
a ³ F ₅ – z ² G ₆ ^o	901.7653	901.5178	332.4506	–	–	–	–	proposed narrow closed cooling line from metastable a5F5

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).